

IKT215 Exercise 2: Bayesian Decision Theory

January 29th, 2026

Exercise sessions, while not mandatory, are an integral part of your learning experience and exam preparation. Some exam questions will be based on tasks covered during these sessions. Although you have the flexibility to complete exercises at home and ask questions later (preferably within one week), I strongly recommend attending the sessions in person for immediate support and guidance. This allows you to benefit from direct interaction and real-time problem-solving. I respond promptly to emails (manuel.s.mathew@uia.no).

Question 1

Educational experts have built a model to predict whether an incoming freshman will pass or fail their introductory calculus class, based on the number of hours/day spent studying the lectures (S) and doing the homework (H). Using a binary random variable Y to code for pass/fail, the model is given by:

$$Y = \begin{cases} 1 \text{ (pass)} & \text{if } S + H + N > 5, \\ 0 \text{ (fail)} & \text{otherwise} \end{cases}$$

for $S, H, N \geq 0$, where N is an unobservable variable corresponding to factors such as motivation, here translated to the equivalent in hours/day. The variables S , H , and N are modelled as independent and exponential with parameter $\lambda = 1$. Compute the Bayes error for the optimal predictor of whether a given student will pass the class based on S and H .

Question 2

A hospital wants to predict whether patients have Disease X based on the available dataset *diseaseX.csv*.

Tasks

- Exploratory analysis.
- Implement a **Gaussian Naive Bayes classifier** to predict disease status using relevant features.
- Visualize the decision boundary and calculate accuracy.

Question 3

A credit card fraud detection systems need to balance fraud detection (true positives) and false alarms (false positives) based on the available dataset *fraud_detection_dataset.csv*.

Tasks

- a) Exploratory analysis (Optional).
- b) Train a classifier and plot its ROC curve.
- c) Calculate the Bayes error rate.
- d) How would you explain the relationship between the ROC curve and Bayes error to a non-technical bank manager?

Question 4

In a high school library section, there are 68 books. Out of these, 31 are fiction and the rest are non-fiction. In this section, there are two major categories: mystery and historical. A catalogue review shows that 21 out of 31 fiction books are categorized as mystery, while only 8 non-fiction books are categorized as mystery. The librarian randomly selects a book from the historical section. What is the probability of the selected book being non-fiction?

Tasks

- a) Calculate the prior probabilities randomly selecting a fiction and a non-fiction.
- b) Write the Bayes' equation for finding the probability of randomly selecting a non-fiction book given that the book is from the historical section.
- c) Find the probability of a randomly selected non-fiction is categorized as historical.
- d) Find the probability of a randomly selected book being one categorized as historical.
- e) Using the above calculated answers, find the probability of randomly selecting a non-fiction book given that the book is from the historical section

Question 5

In a glass bottle, there are marble balls in two colours: red and blue. These marble balls have been manufactured by two companies: A and B. 20% of the marble balls in the glass bottle are from the manufacturer A and 80% are from manufacturer B. The one who collected these marble balls says that 90% of marble balls from manufacturer A are red while 50% of marble balls from the manufacturer B are red. You are asked to randomly take one marble ball out from the bottle. If the marble ball you took out from the bottle was red, what is the probability that the manufacturer of that ball is company A.

Question 6

Create a program to automate Bayesian classification for a two-class problem with one feature. The program must learn from training data, apply the Bayesian decision rule to classify new instances, and validate its performance using an artificially generated dataset.

Tasks

1. Prior Probability Extraction

- Compute prior probabilities $P(a)$ and $P(b)$ from the training data.
- Ensure priors reflect unequal sample sizes (e.g., 200 samples for class 'a', 100 for class 'b').

2. Class-Conditional Probability (Likelihood) Calculation

- Discretize feature values into **bins** (e.g., equal-width intervals).
- For each bin, compute $P(x \in \text{bin} \mid C_i)$ for all classes C_i .

3. Posterior Probability Calculation and Visualization

- For each bin, calculate the posterior probabilities $P(C_i \mid x \in \text{bin})$ using Bayes' theorem.
- Plot posterior probabilities for both classes across all bins (e.g., bar/line plot).

4. Program Validation with Artificial Dataset

Step 1: Dataset Generation

- Generate training data:
 - Class 'a': Sample x from a normal distribution (e.g. $\mathcal{N}(\mu = 25, \sigma^2 = 1)$).
 - Class 'b': Sample x from a normal distribution (e.g. $\mathcal{N}(\mu = 20, \sigma^2 = 1)$).
 - Ensure the number of samples per class is unequal (e.g., 200 samples for class 'a', 100 for class 'b').
- Store the data in a two-column format: one column for x , another for the class label ('a'/'b').

Step 2: Training Phase

- Extract priors $P(a)$ and $P(b)$ from the training data.
- Compute likelihoods $P(x \in \text{bin} \mid C_i)$ using the binned feature values.

Step 3: Posterior Analysis

For each bin, compute and print the posterior probabilities $P(C_i \mid x \in \text{bin})$ in tabular format.

Step 4: Testing and Accuracy Evaluation

- Generate a separate test dataset by sampling new instances from the same distributions (e.g., 50 samples from 'a', 50 from 'b').
- For each test instance, use the Bayesian decision rule (classify to the class with the highest posterior) to predict its label.
- Compare predicted labels with ground-truth labels to compute classification accuracy.